

Sudeera Dilshan

AI/ML Engineer

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About Me

Final-year B.Sc. (Hons) Artificial Intelligence undergraduate at the University of Moratuwa with hands-on industry experience in LLM integrations, agentic AI workflows, and MCP server development. Passionate about Data Science, machine learning research, computer vision, and building production-ready AI systems. Skilled in full-stack and mobile app development, with a proven track record of delivering impactful AI solutions. A published researcher, technical content creator on Medium, and competitive programmer committed to continuous learning and responsible AI innovation.

Education

B.Sc. (Hons) in Artificial Intelligence (CGPA: 3.33) *2022 – 2026*
University of Moratuwa, Colombo

G.C.E. Advanced Level – Common Stream *2017 – 2019*
Richmond College, Galle

- Combined Mathematics – A, Physics – A, Information Technology – B Z-Score: 2.2305

Experience

Associate AI Engineer *Aug 2025 – Oct 2025*
Ariso Technologies (Pvt) Ltd

- Developed and integrated production-level LLM pipelines and MCP server components to power intelligent, context-aware AI services.
- Contributed to time-series forecasting model development and optimised deployed AI systems for scalability and reliability in production environments.
- *Technologies: Python, LLMs, LangGraph, MCP, FastAPI, Docker, Prophet, statsmodels*

AI Engineer Intern *Feb 2025 – Jul 2025*
Ariso Technologies (Pvt) Ltd

- Built and deployed agentic AI workflows with multi-step reasoning and tool-use capabilities, integrating LLMs into real-world business automation pipelines.
- Engineered MCP server modules and AI-powered microservices, enabling seamless orchestration of external data sources and APIs.
- Designed and implemented an agent orchestration framework using Mosquitto message broker for sending commands to agents and Redis for distributed cache memory management.
- Built knowledge base services for LangChain and LangGraph agents, enabling dynamic assignment of relevant knowledge bases to specialised agents for context-aware reasoning.
- Created and managed agent workflows using Langflow, streamlining the visual design and orchestration of multi-agent AI pipelines.
- *Technologies: Python, LangChain, LangGraph, LLMs, MCP, Langflow, Mosquitto, Redis, FastAPI, Docker*

Technical Skills

Languages: Python, Java, R, C, Dart, JavaScript, SQL

GenAI & Agentic Frameworks: LangChain, LangGraph, Langflow, LangSmith, MCP

AI/ML Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, SHAP, HuggingFace

Data Analysis: pandas, numpy, matplotlib, seaborn, Tableau

Data & Vector Stores: MongoDB, MySQL, PostgreSQL, Neo4j, FAISS, ChromaDB, Redis

MLOps, Cloud & Messaging: Docker, AWS, Jenkins, GitHub Actions, Firebase, Git, Mosquitto

Backend & Full-Stack: FastAPI, Flask, NodeJS, ReactJS, Spring Boot, Flutter

Projects

Soft Prompting-Based SLM Generalization - Meta-Learning & Fuzzy Reasoning *GitHub | Final Year Research*

- Conducting research to improve cross-task generalisation in Small Language Models (SLMs) through chain-of-thought knowledge distillation and meta-learning.
- Designed a custom architecture featuring a lightweight MLP with a top fuzzy layer for adaptive soft prompt optimisation, enabling improved reasoning across diverse tasks.

- Implemented a custom transformer model, MLP, and an ANFIS (Adaptive Neuro-Fuzzy Inference System) module, with soft prompts trained using First-Order MAML (FOMAML).
- *Technologies: Python, PyTorch, Qwen2.5-0.5B, Meta-Learning, FOMAML, ANFIS, Fuzzy Logic*

Voicense - AI-Powered Lecture Note Generation Platform *GitHub*

- Developed an AI-powered mobile application for automated lecture recording, transcription, and intelligent note generation; served on the core backend and AI pipeline team.
- Published findings at the 9th International Conference on Information Technology Research (ICITR 2024), IEEE.
- Delivered core features including real-time audio transcription, automatic structured note generation, topic-wise summarization, and an AI chat assistant for querying lecture content.
- *Technologies: Flutter, FastAPI, MongoDB, LangChain, Gemini API, Whisper AI, Docker*

StreamAI - Hybrid Machine Learning Recommender System *GitHub*

- Developed a hybrid recommendation engine utilizing K-Means for user segmentation, SVD (Matrix Factorization) for collaborative filtering, and TF-IDF for content-based filtering.
- Designed a dynamic routing algorithm that transitions users from Popularity-based to Collaborative algorithms as their data profile grows, solving the cold-start problem.
- Packaged the machine learning pipeline into a production-ready, interactive web dashboard using Streamlit.
- *Technologies: Python, scikit-learn, scikit-surprise, Streamlit, Pandas*

Bias-Free Loan Eligibility & Risk Explainability System *GitHub*

- Built an end-to-end ML system for loan eligibility prediction integrating SHAP-based explainability and fairness auditing to ensure transparent, ethical financial decisions.
- Integrated a small language model (Qwen2.5-0.5B) to convert SHAP-based explanations into clear, human-readable narratives for end users.
- *Technologies: Python, XGBoost, SHAP, FairLearn, FastAPI, ReactJS, Docker, Scikit-learn, Qwen2.5-0.5B*

Time Series Transit Forecast: Visualized Analysis *GitHub*

- Automated a data pipeline using the Socrata API to extract and preprocess 20+ years (9,000+ records) of daily public transit data.
- Engineered temporal lag features to train an XGBoost regression model, outperforming a baseline Facebook Prophet model in predicting daily ridership.
- Developed an interactive Streamlit web application with Plotly to visualize historical trends, seasonal anomalies, and run dynamic model forecasts in real-time.
- *Technologies: Python, XGBoost, Prophet, Streamlit, Pandas*

Virtual Mouse Using Hand Gestures *GitHub*

- Developed a hands-free virtual mouse system using real-time webcam-based hand gesture recognition, enabling full cursor control without physical hardware.
- Implemented real-time hand landmark detection and tracking to map finger positions and gestures to cursor movement, left/right clicks, scrolling, and drag actions.
- *Technologies: Python, OpenCV, MediaPipe, PyAutoGUI*

Publications

- **Voicense: AI-Powered Lecture Note Generation Tool**
  9th International Conference on Information Technology Research (ICITR) 2024, pp. 1–6, IEEE, Dec 2024. University of Moratuwa, Sri Lanka. DOI: 10.1109/ICITR64794.2024.10857774

Certifications

- Advanced Python Programming – Source Code
- MCP: Build Rich-Context AI Apps with Anthropic – DeepLearning.AI
- LangChain for LLM Application Development – DeepLearning.AI
- Neo4j Certified Professional – Neo4j GraphAcademy
- Kubernetes From 10000 Feet – CoDeKu DevOps Academy

Achievements & Extracurricular Activities

- **6th Place**, CodeRush 2023 (Organised by INTECS)
- **Participant**, ACES Coders V10.0 2023 – University of Peradeniya
- University of Moratuwa Karate Team Representative – 2022, 2023
- Technical Content Creator on Medium – @dilshanrgs31

References

Dr. (Ms.) I. Thilini S. Piyatilake
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